

Hospital Bed & Resource Management System

Advanced Excel-Based Healthcare Operations Solution



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SWIPE →

Objective:

Built a centralized hospital management system in Excel to automate:

- Patient Admission
- Bed Allocation
- Doctor Assignment
- Billing & Insurance
- Hospital Analytics Dashboard

Tools & Technologies Used:

- Microsoft Excel
- VBA Macros
- Pivot Tables & Charts
- Slicers & Timeline
- Conditional Formatting
- XLOOKUP, IF, COUNTIF, SUM



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Real Problems in Traditional Hospital Operations

- Manual patient admission process
- No real-time bed tracking
- Uneven doctor workload
- Delayed billing calculations
- Poor operational visibility
- No centralized analytics system

Need for the Solution:

To create:

- faster hospital workflows
- automated resource management
- real-time monitoring
- centralized operational analytics



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Excel System Architecture Developed:

- ✓ Master_Data
- ✓ Patient_Form
- ✓ Patient_Data
- ✓ Bed_Master
- ✓ Allocation_Engine
- ✓ Doctor_Master
- ✓ Billing_Logic
- ✓ Billing_Master
- ✓ Insurance_Master
- ✓ Hospital_Dashboard

Workflow

Patient Admission → Bed Allocation → Doctor Assignment → Billing → Dashboard Analytics



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Key Dynamic Features

Smart Bed Allocation

- Department-wise matching
- Bed type matching
- Availability tracking

Dynamic Doctor Assignment

- Least workload doctor selection
- Capacity-based assignment
- Auto busy/available status

Dynamic Billing Engine

- Stay-based billing
- ICU & emergency charges
- Insurance deduction logic

Executive Alerts

- Low bed availability
- ICU overload alerts
- Pending allocation warnings



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Executive Dashboard & Analytics

Dashboard Features - KPI Cards

- Total Patients
- Occupied Beds
- Available Beds
- Busy Doctors
- ICU Patients
- Emergency Cases

Interactive Analytics

- Department-wise patient analysis
- Bed occupancy analysis
- Doctor workload monitoring
- Daily admission trends

Interactive Controls

- Slicers
- Timeline Filters
- Dynamic Alerts

Business Impact & Conclusion

- ✓ Faster patient admission workflow
- ✓ Improved bed utilization
- ✓ Balanced doctor workload
- ✓ Automated billing process
- ✓ Real-time operational visibility
- ✓ Better management decision-making



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Key Learning Outcomes

This project demonstrates:

- ✓ Advanced Excel Automation
- ✓ VBA Logic Building
- ✓ Dashboard Development
- ✓ Business Analytics Thinking
- ✓ Workflow Optimization
- ✓ Real-World Problem Solving

Final Conclusion

Successfully designed an end-to-end hospital operations & analytics system using Excel that simulates real-world healthcare ERP workflows with automation, analytics, and dynamic reporting



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